

An Auditable Market-Microstructure Framework for Spoofing-Compatible Sequence Detection

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Abstract

How can surveillance data isolate order-book sequences that are compatible with spoofing without treating ordinary liquidity provision as manipulation? We address this question with an auditable microstructure framework that reconstructs the visible limit order book and groups related execution messages into a single analytical execution. An execution is anchored either by a [passive execution of the actor's resting order](#) or by the actor's aggressive execution. We attribute orders through the original client code when it is available and otherwise through the firm identifier, while preserving the identity level. For each execution, the candidate set comprises [orders placed by the same actor on the opposite side](#) that are active in the top 10 ranks immediately before the first execution and were first observed no more than 90 seconds earlier. The strict gate passes only when four conditions hold simultaneously: cancellation of at least one candidate order is canonically assigned within 2 seconds; attributed withdrawal exceeds execution quantity; [the observed favorable movement in the midprice before execution](#) is positive; and [the observed midprice reversion measured from the cancellation](#) is positive over a 2-second horizon. An empirical depth kernel weights visible liquidity by book rank. In the Risanamento, Nexi, and Ferrari samples, respectively, the reconstruction identifies 205, 1,684, and 2,475 matched executions, of which 12, 231, and 370 satisfy the strict gate. Both passive and aggressive execution routes contribute to these sets. An exploratory comparison with 36 timed external alert records yields 34 distinct subject-instrument intervals after overlapping or endpoint-sharing records are merged. Every interval contains a recovered execution, 30 contain an attributed withdrawal, and four contain a strict sequence. The four strict overlaps occur only at firm scope, where the external firm maps to multiple canonical actors. These findings support reproducible surveillance triage, but they do not constitute a statistical test, identify manipulative intent, estimate causal effects, measure spoofing prevalence, or validate recall against independently adjudicated manipulation labels. Legitimate liquidity provision remains an explicit alternative explanation.

Keywords. Spoofing detection; market manipulation; limit order book; high-frequency trading; anomaly detection; market maker; market microstructure

1 Introduction

Electronic continuous double auctions organize price discovery through a limit order book (LOB). The LOB records the visible bid and ask orders submitted by market participants and changes whenever an order is entered, amended, executed, or cancelled. Algorithmic and high-frequency trading have increased the speed of these updates and can improve short-term liquidity. The same speed permits large displayed positions to be submitted and withdrawn over intervals that can be difficult to assess as complete sequences in real time.

Spoofing exploits this sequencing problem. In a stylized account, a participant submits a large order on one side of the LOB that is not intended to execute. The displayed quantity may change observed imbalance or microprice indicators and may prompt other participants to revise quotes or submit aggressive orders. The participant can then execute a smaller order on the opposite side at a more favorable price and withdraw the large displayed order. The Market Abuse Regulation (MAR) prohibits orders and behavior that create false or misleading signals about the supply of, demand for, or price of a financial instrument [3].

The observable actions in this mechanism are not unique to spoofing. Liquidity providers also submit, amend, and cancel orders at high frequency, and they may have high order-to-trade ratios. They remove quotes to update prices, control inventory, or limit adverse selection. Static size thresholds, cancellation ratios, and volumetric triggers therefore cannot by themselves distinguish manipulative conduct from legitimate liquidity provision. Such rules can produce false positives and cannot establish the participant’s intent.

A deliberately broad diagnostic makes this overlap visible. Figure 1 ranks actors by the rate at which they cancel liquidity on the side opposite to an execution within the next two seconds. Positive rates occur in all three samples, but the frequency and the number of supporting executions vary substantially across actors. The statistic is deliberately agnostic about why an order was cancelled. A high value can arise from ordinary quote and inventory management, so it is not evidence of manipulation. The dispersion instead motivates a reconstruction that asks which rapid cancellations also belong to the more specific sequence implied by the hypothesized mechanism.

This paper asks whether the observable conjunction implied by the hypothesized mechanism can be reconstructed from messages attributed to individual participants without treating frequent cancellation itself as evidence of spoofing. Measures of the actor’s complete resting profile over the top 10 ranks first describe the shape of an actor’s displayed liquidity; these diagnostics do not determine eligibility. For each passive or aggressive execution, we then fix a candidate set containing orders placed by the same actor on the opposite side that are active in the top 10 ranks and no more than 90 seconds old. A matched execution has at least one cancellation of a candidate order canonically assigned within 2 seconds. The strict gate with four conditions additionally requires the attributed withdrawal to exceed execution quantity, the observed favorable movement in the midprice before execution to be positive, and the observed midprice reversion measured from the cancellation to be positive over the 2-second reversion horizon. Actor identity is a separate data problem, so we use the original client code when present and the firm identifier only as a fallback. Each physical cancellation is assigned to at most one execution.

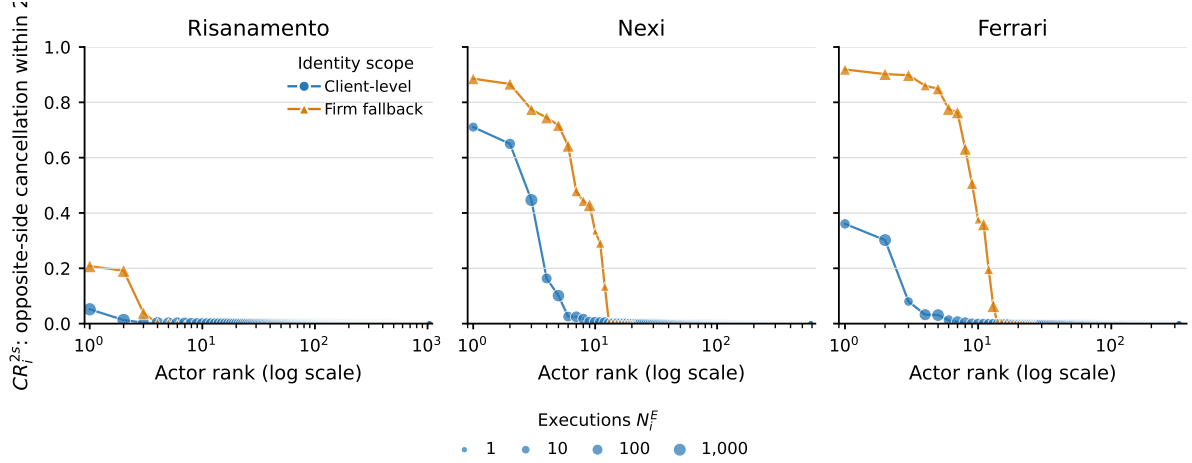


Figure 1: Rate at which each actor cancels opposite-side liquidity within two seconds of its own execution. For actor i , the denominator N_i^E contains all of its passive and aggressive executions. The numerator counts executions followed by at least one cancellation by the same canonical actor on the opposite side during $(t_c^{\text{end}}, t_c^{\text{end}} + 2 \text{ s}]$. The cancellation must also follow the final constituent execution message in canonical source order and remain in the same trading partition. Each execution contributes at most one unit to the numerator, regardless of how many qualifying cancellations follow it. Each execution defines its own window, so the same cancellation can follow more than one nearby execution. Client-level and firm-fallback actors form two separate curves and are ranked from highest to lowest rate within each identity scope and instrument. The actor-rank axis is logarithmic, and no actor identifier is displayed. Identity follows the client-first rule: the original client code is used when available, with the firm identifier used only as a labelled fallback. Marker area increases with $\log(1 + N_i^E)$ to make sample support visible; marker shape and color also distinguish the two identity scopes. No actor is removed for low activity. Firm fallback is a coarser scope that can pool client activity, so its rates are not directly equivalent to client-level rates. This broad statistic does not impose the later conditions on candidate age, the fixed candidate set, withdrawal size, or the price path, and it does not use WMSCI, MSCl, or the strict gate. A high rate is not evidence of manipulation.

The empirical reconstruction identifies [rapid withdrawals by the same actor](#) in all three samples and after both execution routes. The full four-condition sequence is less common because it additionally requires attributed withdrawal to exceed execution quantity, favorable movement, and reversion. This distinction is the main interpretive result: a mechanical withdrawal can motivate review, but it is not equivalent to a sequence that also satisfies [the conditions on size and on the price path](#) of the stylized mechanism. The output is therefore an auditable set of descriptive screening diagnostics for analyst review, not a statistical validation, a label of manipulation, or an estimate of causal price impact. Within the candidate set, the Withdrawal Multilevel Spoofing Conditionality Index (WMSCI) is [the primary metric used to prioritize events](#). It ranks events by the scale and speed of attributed withdrawal relative to the execution, but it does not define the four-condition gate.

An exploratory comparison with externally supplied surveillance alerts gives the same evidence hierarchy. The 36 timed source records form 34 distinct subject–instrument intervals after overlapping or endpoint-sharing records are merged. All 34 intervals contain a recovered execution, 30 contain an attributed withdrawal, and four contain a strict sequence. All four strict overlaps occur where [the external subject, identified at firm level, maps to multiple canonical actors](#), so they establish [coverage at subject scope rather than correspondence with an exact actor](#). Because the external alerts are positive seeds rather than independently adjudicated labels, this comparison reports recovery relative to the source alerts. [Strict recall for an exact actor](#) is defined only for intervals whose external and internal identities have aligned granularity; it is not recall against confirmed manipulation.

2 Market Making Versus Spoofing

The screening model starts from a contrast between two stylized economic benchmarks. Both can generate many order messages and cancellations, but they assign different hypothesized roles to displayed liquidity. This contrast does not imply that every market maker quotes symmetrically or that every spoofing episode follows one fixed pattern. It identifies observable features that can guide surveillance.

2.1 The Market Maker’s Problem

A market maker supplies liquidity by posting resting orders on the bid and ask sides of the book. In [a stylized benchmark based on inventory risk](#), the dealer seeks to earn the bid–ask spread while controlling inventory risk and exposure to adverse selection. Optimal quotes therefore depend on both [rates of order arrival](#) and the dealer’s inventory.

[Models of stochastic control](#) often express this problem through a Hamilton–Jacobi–Bellman equation. The dealer chooses bid and ask quotes that balance execution revenue against inventory risk. For example, a long inventory exposes the dealer to a price decline. The dealer can respond by shifting both quotes downward: a lower ask encourages purchases from the dealer, while a lower bid discourages further sales to the dealer. The quote adjustment depends on the midprice, the spread required for compensation, and the penalty for inventory away from the target.

Within this benchmark, frequent cancellation is [an action for risk management](#). The market maker cancels and reposts orders as the midprice changes, as [the risk of adverse selection](#) changes, or as inventory moves away from its target. The dealer generally maintains [a presence on both sides](#) near the top of the book within this stylized benchmark, although actual quoting can become asymmetric because of inventory, information, or market conditions. Expected profits in the benchmark come primarily from spread capture rather than from deliberately inducing a directional change in the midprice.

2.2 The Spoofer’s Problem

In the stylized spoofing mechanism, the participant is hypothesized to seek to improve the price of a directional execution. The participant displays a large order on one side of the book and submits a smaller order on the opposite side. The large displayed order creates an asymmetric depth profile. Algorithms or traders that use bid–ask volume imbalance may interpret this profile as evidence of [pressure on prices over a short horizon](#) and adjust their orders. If other traders respond to the displayed asymmetry, the resulting price movement could improve the execution of the smaller opposite-side order.

The displayed order creates a trade-off between influence and execution risk. An order near the best quote is prominent in the book but is also more likely to execute. An order posted at Level 2, Level 3, or deeper has a buffer against immediate execution while remaining visible in [order-book data across multiple levels](#). A participant implementing the stylized strategy may therefore distribute the large profile across deeper levels rather than place all of it at the touch.

After the smaller order executes, rapid withdrawal of [the displayed profile on the opposite side](#) is consistent with reducing subsequent execution exposure under the hypothesized strategy. This timing motivates the diagnostic contrast with ordinary quote updates: the candidate withdrawal follows [the execution on the opposite side](#) rather than only changes in fair value, inventory, or [the risk of adverse selection](#). [Table 1](#) summarizes these stylized incentives; none is directly observed as intent.

3 Existing Approaches

[Methods for detecting spoofing](#) differ mainly in the object they classify. [Methods based on rules reconstruct a stylized episode from the mechanics of the order book](#). Statistical methods monitor unusual states of order flow. Sequence and graph models learn temporal or relational patterns from event data. For the present setting, this taxonomy is useful because it distinguishes methods that reconstruct attributable order sequences from methods that detect unusual states or learned temporal patterns. [Figure 2](#) summarizes the method families, which can overlap in practice.

Early empirical studies operationalize [screens for spoofing surveillance through interpretable conditions from the order book](#). Using [Korea Exchange data at the account level](#), [7] identify large orders placed far from the best quote, followed by opposite-side trading and subsequent withdrawal. Related surveillance rules have been applied to Brazilian equities [9], FX spot markets [11], cross-market FX activity [10], and Taiwan derivatives [1]. These studies con-

Table 1: Stylized differences used to construct the surveillance diagnostics. These differences describe economic benchmarks, not observed intent or necessary conditions for either activity.

Microstructure Feature	Legitimate Market Maker	Hypothesized Spoofing-Compatible Mechanism
Primary objective	Earn the bid–ask spread while controlling inventory and the risk of adverse selection .	Improve a directional execution through a displayed order-book signal.
Order-book imbalance	Usually maintains two-sided liquidity, with asymmetry arising from inventory and risk management.	A large, short-lived asymmetry forms the candidate posture.
Cancellation motive	Update quotes to track fair value, control inventory, or limit adverse selection.	Withdrawal after the smaller opposite-side execution is consistent with the mechanism.
Posting-distance trade-off	Quote near the touch to increase execution probability and earn the spread.	Remain visible while reducing the risk that the displayed posture executes.
Source of expected gain	Spread capture, without intentionally moving the midprice in a chosen direction.	A potential favorable price change for the smaller order on the opposite side.

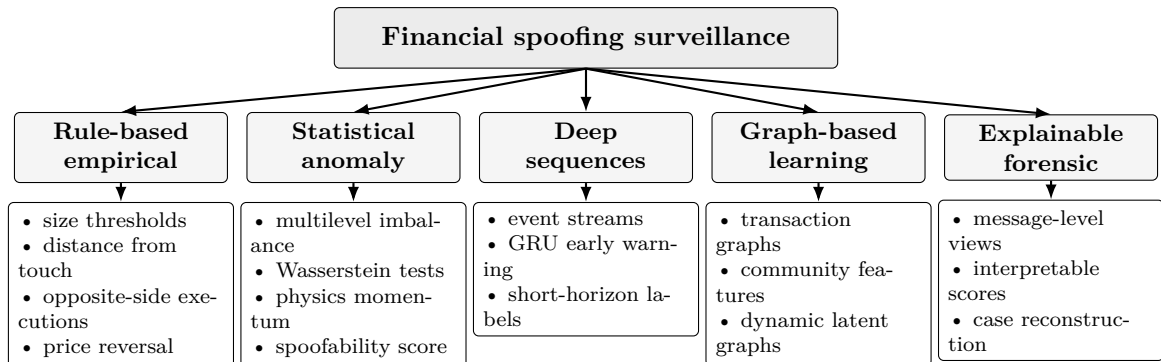


Figure 2: Method families in [the literature on spoofing detection](#). The categories can overlap; each column identifies the main object or representation used by a strand of work.

nect a screen to observable features such as order size, posting distance, cancellation timing, opposite-side execution, and short-lived price impact. However, the reported implementations use [thresholds specific to each venue](#), and their reconstruction can require privileged message data or account identifiers.

A second line of work monitors statistical anomalies in the order book. [13] derive spoofing incentives from multilevel imbalance and propose [a statistic based on Wasserstein distance for monitoring in real time](#). [8] represent the order book as [a system described using concepts from statistical physics](#) and detect abnormal momentum in cryptocurrency markets. Explainable forensic methods partly overlap with this family. [Visualization at the message level](#) can reconstruct prosecuted episodes [2]. Interpretable probabilistic neural networks instead estimate how an isolated limit order changes the conditional distribution of future prices and the expected gain from a hypothetical manipulation strategy in centralized cryptocurrency markets [4]. The latter object measures [the spoofability of an individual order](#); it does not establish whether one participant generated the candidate posture, [the smaller order on the execution side](#), and the withdrawal. Our framework addresses this complementary attribution problem by linking these observed actions through canonical actor keys while recording whether each linkage is at client or firm scope.

Machine-learning methods add temporal or relational structure. [14] formulate spoofing as [a problem of early warning](#) and train a gated recurrent unit model on [streams of events from cryptocurrency order books](#). [Methods based on graphs](#) move the unit of analysis from isolated orders to networks of traders and transactions. This representation can capture coordinated spoofing patterns [5, 6, 15]. Reported models combine temporal transaction graphs, community features, and dynamic latent representations. Their practical performance remains difficult to compare because available labels, market settings, and validation designs differ.

These studies imply two requirements for the present surveillance setting. First, the features must retain their microstructure interpretation. The screen must record candidate volume, posting distance from the touch, cancellation timing, and the relation to an opposite-side execution. Second, any eventual validation must reflect scarce and noisy enforcement labels, rare positive cases, and the analyst cost of false positives. Given the evidence currently available, the model below preserves the auditability of rule-based surveillance while producing [descriptive order sequences attributed to individual participants](#) from proprietary event data.

4 Data

We use [three samples of CONSOB messages from the limit order book](#). The Risanamento sample covers June–November 2024, the Nexi sample covers 23 April 2024, and the Ferrari sample covers 13 June 2024. These samples differ in duration and participant composition, so [differences across instruments](#) are descriptive rather than controlled comparisons.

For an exploratory coverage audit, we also use an externally supplied CONSOB surveillance report containing 37 alert records supplied as positive seeds. These externally identified periods are used to test whether the reconstruction covers the reported activity; they are not independently adjudicated manipulation labels. The source contains 35 timed Ferrari records, one timed Nexi record, and one Risanamento record that identifies only the trading date. We

preserve every source record and merge timed intervals that overlap or share an endpoint only within the same instrument and external subject. This procedure yields 34 distinct timed intervals, while [the Risanamento record that identifies only the date remains outside the denominator for event-level analysis](#).

For timed records, we parse the source bounds at millisecond precision and include both endpoints. To place raw events on [a common axis of event time](#), we use the first nonmissing timestamp in [the priority order of the source fields](#) TRADETIME, BOOKOUTTIME, BOOKIN, and SEQUENCE TIME. We recover an execution when at least one of its constituent execution messages lies inside the closed union interval; [we then read whether the execution has a matched withdrawal and satisfies the strict gate from its diagnostics at the event level](#). The union interval, rather than each potentially overlapping source record, is therefore the observational unit for timed counts.

We match each external subject in the identity namespace reported by the source. Consequently, an external firm can cover several canonical actor keys when client identifiers are available internally. We therefore distinguish [overlap at subject scope from correspondence with an exact actor](#) and treat the latter as identifiable only when the external identity maps to one canonical actor at the same granularity. The alert report contains no negative sample, so the comparison cannot estimate [a rate of false positives](#).

For each instrument, we reconstruct the visible order book from order-level events and infer the tick size from the reconstructed best quotes. [We compute diagnostics for each actor's resting profile over the first ten book ranks and separately construct a set of candidate orders on the opposite side that satisfy the age restriction. The main specification uses a 10-second window after execution for structural reconstruction, a 2-second window for attributing rapid withdrawals, a 2-second horizon for reversion measured from each assigned cancellation, and a maximum age of 90 seconds for candidate orders.](#)

The original client code is not available on every message. We therefore define one canonical actor key before constructing events. If the original client code is present, it defines the actor; only when it is missing do we use the firm identifier. The technical client value 0 is treated as missing rather than as an actor, so a message carrying this sentinel uses the firm fallback when a firm identifier is available. The two namespaces are kept distinct, and every output records whether attribution is at the client or firm level. [A key derived from the client code and a fallback key based on the firm identifier are distinct canonical identities; the fallback does not merge a message identified by client code with a message identified only at firm level.](#) Messages missing both identifiers are excluded from [analysis that requires actor attribution](#); none occurs in the reported samples. Firm fallback is coarser than client attribution and can combine several clients represented by the same firm. It is therefore an identity fallback, not evidence that those clients are one economic account.

Execution role is defined independently of actor identity. We retain individual execution messages marked exclusively passive or exclusively aggressive and reject messages for which the two role flags are both set or both absent. Related execution messages are grouped into one analytical execution when they share the trading partition—trading date, venue, market, instrument, and EMM code—as well as canonical actor, order identifier, execution role, and side.

Passive executions also require a common resting price; aggressive executions can span several traded prices under the same order and sweep. Consecutive constituent execution messages must be no more than 100 milliseconds apart, with no intervening cancellation, modification, or lifecycle event that creates a new order for the same order. The 100-millisecond threshold is an operational choice, and the reported findings are conditional on it; sensitivity to alternative gaps between execution messages is not reported here. Identifiers of raw messages and the constituent messages assigned to each execution remain available as provenance, while executions form the analytical unit for event metrics and aggregation by actor and partition. The resulting surveillance dossiers support analyst review and are not enforcement labels.

5 Model

The model translates the stylized contrast between market making and spoofing into a sequence of observable conditions. Its central question is whether recent opposite-side orders disappear specifically after the same participant executes. To answer it, we construct two distinct objects. The first is the participant’s complete resting profile across the top n book levels, which supplies descriptive DWI, SCI, and MSCI shape diagnostics. The second is a set of candidate orders on the opposite side that satisfy the age restriction, which supplies quantities for matched withdrawal and WMSCI, the metric used to prioritize events. Separately, the strict gate applies the prespecified conditions on withdrawal size, favorable movement, and reversion. These sign and size restrictions are screening conditions, not estimates of causal influence or intent. The quantities describing the resting profile, withdrawal, and price response remain separate because they describe different parts of the mechanism.

5.1 A Spatio-Temporal Framework for Spoofing Detection

Figure 3 sketches the hypothesized sequence. The detector later translates each state into observable order, execution, and cancellation records rather than inferring the participant’s purpose.

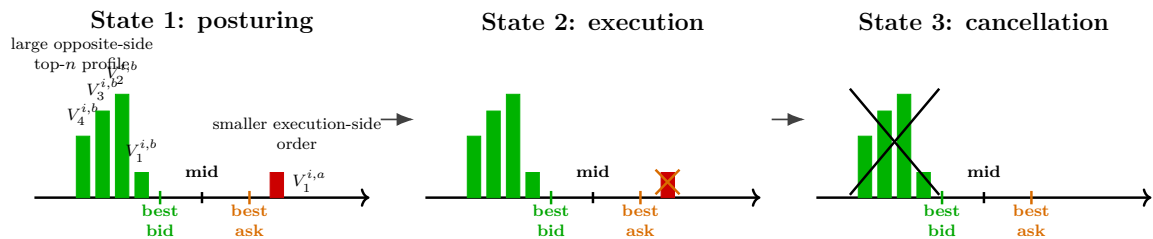


Figure 3: Schematic top- n timeline for the hypothesized spoofing mechanism. The large candidate posture forms a depth profile for the agent over several bid levels, while the smaller order rests near the opposite best quote. The construction applies symmetrically to candidate episodes with buy or sell intent. The figure shows the passive execution route; the operational reconstruction also admits an aggressive execution by the same actor.

The method represents a candidate episode as a conditional change in the participant’s LOB footprint. A single order at the best bid or ask does not capture this change. We instead use the participant’s liquidity profile over the first n price levels. This representation includes postures

placed at Level 2, Level 3, or across adjacent levels, where displayed liquidity can remain visible while facing less immediate execution risk.

The hypothesized mechanism for participant i has three states:

1. **State 1 (Candidate Posturing):** Shortly before the execution, the participant displays a large liquidity profile on one side of the book, typically away from the touch. The actor has [an execution anchor on the opposite side](#), which may be passive or aggressive in the operational reconstruction.
2. **State 2 (Execution):** The large profile is present when the smaller opposite-side order executes. In the hypothesized mechanism, the profile can coincide with a more extreme measured imbalance before this execution.
3. **State 3 (Withdrawal):** The participant rapidly cancels the opposite-side profile after the execution; under the stylized mechanism, this timing is consistent with reducing subsequent execution exposure.

State 2 has two observable routes. In the passive route, the actor's resting order executes. In the aggressive route, the actor initiates the trade against resting liquidity. The role flag changes how execution price and quantity are read and how individual execution messages are grouped; it does not change actor identity. In both routes, the candidate posture is sought among the same actor's recent orders on the side opposite to the actor's execution side. An aggressive anchor is therefore not itself evidence of spoofing, just as a passive anchor is not evidence of legitimate market making.

These three states illustrate the hypothesized mechanism; their qualitative descriptors do not define [membership in the strict set](#). For either execution route, the candidate set consists of [orders placed by the same actor on the opposite side](#) observed in the top 10 ranks before the execution and satisfying the 90-second age limit. A cancellation within 2 seconds supplies [rapid evidence of a matched withdrawal](#). The strict gate then requires attributed withdrawal to exceed execution quantity and both midprice response diagnostics to be positive. The detector observes these conditions, not the participant's intent or the profile's causal effect on other traders. The recency restriction prevents [liquidity that has been resting for a long time](#) from entering a candidate episode solely because it was already resting in the book.

5.1.1 Top- n Participant-Specific Depth Profiles

We first define the [depth profile for each participant](#) used throughout the model. Let $s \in \{b, a\}$ denote the bid or ask side, and let $k = 1, \dots, n$ denote book rank. Rank $k = 1$ is the best quote on side s , and $V_k^{i,s}(t)$ is participant i 's displayed volume at time t on that side and rank. The vector

$$\mathbf{V}_n^{i,s}(t) = \left(V_1^{i,s}(t), \dots, V_n^{i,s}(t) \right) \quad (1)$$

collects these quantities over the first n ranks. The vector preserves where liquidity is displayed rather than reducing the participant's entire side of the book to one total. This distinction matters because a large order near the touch has different execution risk from the same quantity posted at a deeper rank.

Raw displayed volume may not be comparable across assets or intraday periods with different market depth. The relative profile therefore divides participant volume by total displayed market depth at the same side and rank:

$$\tilde{V}_k^{i,s}(t) = \begin{cases} \frac{V_k^{i,s}(t)}{D_k^s(t)}, & D_k^s(t) > 0, \\ 0, & D_k^s(t) = 0, \end{cases} \quad (2)$$

where $D_k^s(t)$ is the total displayed market depth on side s and rank k . The branch for zero depth makes the convention at each rank explicit; an imbalance remains undefined when the participant has no positive weighted footprint on either side. We denote the distance from the touch by $d_k^s(t)$. In tick units, this distance equals the number of ticks between rank k and the best quote on the same side, shifted upward by one. The shift gives Rank 1 a positive distance and preserves the ordering of deeper ranks.

5.1.2 Multilevel Distance-Weighted Imbalance

Economic intuition and a stylized parametric benchmark. We use a depth kernel to reduce the top- n vector to a scalar measure of weighted liquidity. The kernel is designed to assign high weight to ranks that remain represented in multilevel book data while facing comparatively low immediate execution risk. The following parametric kernel illustrates this trade-off:

$$\omega_k^{s,\text{par}}(t) = \frac{e^{-\lambda d_k^s(t)} (1 - e^{-\kappa d_k^s(t)})}{\sum_{\ell=1}^n e^{-\lambda d_\ell^s(t)} (1 - e^{-\kappa d_\ell^s(t)})}, \quad (3)$$

The visibility component $e^{-\lambda d}$ decreases with depth. The protection component $1 - e^{-\kappa d}$ increases as the displayed price moves away from immediate execution. Their product penalizes both the high execution risk near the touch and the lower visibility deeper in the book. The parameters λ and κ control the two shapes. This expression is only a conceptual benchmark: the empirical analysis uses neither these weights nor fitted values of λ and κ . Instead, it constructs empirical proxies at each rank for each instrument and book side.

Figure 4 illustrates the benchmark trade-off and its discrete normalization; it is not an estimated empirical profile.

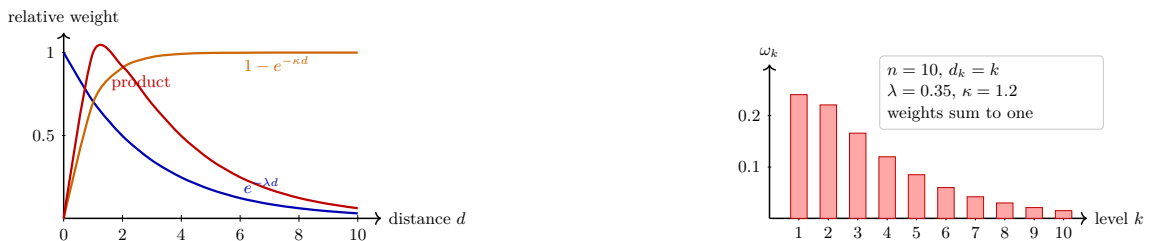


Figure 4: Conceptual depth kernel in equation (3). The left panel shows decreasing visibility, increasing protection from immediate execution, and their product. The right panel shows normalized weights for one illustrative parameter choice. The empirical analysis uses the instrument- and side-specific kernel in equation (10), not these parametric weights.

Operational empirical depth kernel. The operational kernel replaces both exponential components with [empirical quantities defined at each rank](#). We calibrate each book rank k separately for instrument m and side s . A rank receives more weight when its depth has [a strong association with midprice changes over a short horizon](#) and its displayed price is rarely reached over the same horizon. Tick distance can vary for a given rank over time, but the calibrated unit remains the rank.

Operational calculation. We reconstruct [a post-event top- \$n\$ snapshot after every ordered message](#). Let $\mathcal{O}_{m,s,k}$ contain the snapshots in which rank k exists on side s and has positive displayed depth. For each $t \in \mathcal{O}_{m,s,k}$, the indicator $R_{t,k}^{m,s}(H)$ equals one when at least one execution during $(t, t+H]$ occurs at or through the rank's displayed price. An ask rank is reached when the maximum ask execution price in the window is at least the displayed price. A bid rank is reached when the minimum bid execution price is at most the displayed price. We estimate [the hit frequency at each rank across message snapshots](#), denoted $\hat{p}_k^{m,s}(H)$, from pooled hit and exposure counts:

$$\hat{p}_k^{m,s}(H) = \frac{\sum_{t \in \mathcal{O}_{m,s,k}} R_{t,k}^{m,s}(H)}{|\mathcal{O}_{m,s,k}|}. \quad (4)$$

Each observable snapshot contributes once, even when the same rank appears at different tick distances. Thus, [equation \(4\)](#) divides total hits by total exposures; it does not average [hit rates computed separately by distance](#). Realized tick distance is retained only as descriptive metadata and does not define additional kernel rows or enter a parametric fit. Its probability interpretation is conditional on [the sampling scheme based on message snapshots](#). The estimator is [a proxy over a short horizon](#) for whether the displayed price is reached, not an individual order's execution probability, which also depends on queue position, order size, and intervening cancellations. We define the protection component as the complement of the hit probability:

$$\hat{\rho}_k^{m,s}(H) = 1 - \hat{p}_k^{m,s}(H), \quad (5)$$

A value of $\hat{\rho}_k^{m,s}(H)$ near one indicates that rank k is rarely reached over horizon H .

The second component measures the association between displayed depth and [subsequent price changes over a short horizon](#). Let $D_k^{m,s}(t)$ denote total market depth, rather than [depth displayed by participant \$i\$](#) , at rank k . [The total displayed depth across the top \$n\$ ranks](#) at snapshot t is

$$D_n^m(t) = \sum_{r=1}^n \left(D_r^{m,a}(t) + D_r^{m,b}(t) \right). \quad (6)$$

We then define the signed and normalized contribution of side s and rank k :

$$Z_{t,k}^{m,s} = \sigma_s \frac{D_k^{m,s}(t)}{D_n^m(t)}, \quad \sigma_s = \begin{cases} +1, & s = ask, \\ -1, & s = bid. \end{cases} \quad (7)$$

The magnitude $|Z_{t,k}^{m,s}|$ is the fraction of visible top- n depth supplied by that side and rank. The sign makes ask and bid depth contribute in opposite directions to imbalance.

For each snapshot, $M_{t,H}^m$ is the last finite reconstructed midprice observed after t and no

later than $t + H$. The corresponding price change is $\Delta M_{t,H}^m = M_{t,H}^m - M^m(t)$. Let $\mathcal{V}_{m,s,k}(H)$ contain the rank- k snapshots for which both midprices are finite. For [each cell defined by instrument, side, and rank](#), we compute the following pooled empirical covariance in absolute value, normalized by the number of eligible snapshots. The overbars denote sample means over $\mathcal{V}_{m,s,k}(H)$:

$$\hat{\nu}_k^{m,s}(H) = \left| \frac{1}{|\mathcal{V}_{m,s,k}(H)|} \sum_{t \in \mathcal{V}_{m,s,k}(H)} \left(Z_{t,k}^{m,s} - \bar{Z}_k^{m,s} \right) \left(\Delta M_{t,H}^m - \bar{\Delta M}_{k,H}^{m,s} \right) \right|. \quad (8)$$

The absolute value makes $\hat{\nu}_k^{m,s}(H)$ a measure of the magnitude, rather than the direction, of [co-movement specific to that rank](#). We pool observations across realized tick distances before computing the covariance. We do not compute or average separate [covariances for separate distance buckets](#). This quantity is a weighting proxy, not a causal estimate of price impact. It is also expressed in [the price units of the instrument](#). We therefore use it only to allocate weights across ranks on the same side of one instrument, not to compare covariance magnitudes across instruments.

The empirical kernel follows the same multiplicative decomposition as [equation \(3\)](#), but no exponential curve is fitted. Before normalization, we enforce a zero lower bound on protection and apply a named 10^{-12} numerical floor to the absolute covariance:

$$\hat{\rho}_k^{m,s}(H) = \max\{1 - \hat{p}_k^{m,s}(H), 0\}, \quad \hat{\nu}_k^{m,s}(H) = \max\{\hat{\nu}_k^{m,s}(H), 10^{-12}\}, \quad (9)$$

We normalize the product of these two quantities within each instrument and side:

$$\hat{\omega}_k^{m,s}(H) = \frac{\hat{\nu}_k^{m,s}(H) \hat{\rho}_k^{m,s}(H)}{\sum_{\ell=1}^n \hat{\nu}_\ell^{m,s}(H) \hat{\rho}_\ell^{m,s}(H)}. \quad (10)$$

The denominator must be finite and strictly positive. If every rank on a side has zero protection, or if the product is otherwise degenerate, calibration fails instead of returning an uninterpretable all-zero vector. The covariance floor is a numerical regularizer rather than empirical evidence of price association. When estimated covariances are at or below the floor, relative weights can be driven primarily by protection and must not be interpreted as evidence of [price association at a given rank](#). Every reported kernel in [table 2](#) passes the normalization condition and sums to one before rounding.

In the remaining formulas, we use the shorthand $\omega_k^s(t) \equiv \hat{\omega}_k^{m,s}(H)$ for an event in instrument m ; the time argument indicates that the fixed weight is applied to [a profile that varies over time](#). Every event for instrument m uses the same calibrated kernel. The kernel is not re-estimated by actor, event, or observed outcome, and calibration admits only one weight for each (m, s, k) cell. The descriptive analysis uses the full available sample for each instrument. A prospective or out-of-sample application must instead estimate the kernel on an earlier training period and hold it fixed, so that future order flow does not enter the weights.

The operational calibration uses the first ten ranks and the same $H = 10$ second horizon for every instrument. It produces 20 weights per instrument, [one for each combination of side and rank](#). The kernel remains side-specific. To draw one curve per instrument for visualization only,

Table 2: Empirical weights of the top-10 depth kernel, estimated on the full sample at horizon $H = 10$ seconds. Values are rounded to three decimal places and sum to one within each instrument and side before rounding. These weights, rather than fitted exponential parameters, enter the empirical spoofing metrics.

Instrument	Side	$\hat{\omega}_1$	$\hat{\omega}_2$	$\hat{\omega}_3$	$\hat{\omega}_4$	$\hat{\omega}_5$	$\hat{\omega}_6$	$\hat{\omega}_7$	$\hat{\omega}_8$	$\hat{\omega}_9$	$\hat{\omega}_{10}$
Risanamento	Ask	0.127	0.248	0.348	0.062	0.004	0.039	0.018	0.012	0.078	0.064
	Bid	0.526	0.012	0.019	0.146	0.091	0.065	0.082	0.034	0.002	0.024
Nexi	Ask	0.028	0.069	0.033	0.138	0.225	0.149	0.275	0.011	0.020	0.051
	Bid	0.045	0.166	0.184	0.276	0.165	0.033	0.006	0.052	0.064	0.008
Ferrari	Ask	0.095	0.022	0.016	0.209	0.270	0.032	0.144	0.032	0.033	0.147
	Bid	0.148	0.107	0.007	0.054	0.072	0.197	0.102	0.038	0.176	0.100

we average the bid and ask weights at each rank:

$$\bar{\omega}_k^m(H) = \frac{1}{2} \left(\hat{\omega}_k^{m,a}(H) + \hat{\omega}_k^{m,b}(H) \right). \quad (11)$$

Each side-specific kernel sums to one, so $\sum_{k=1}^{10} \bar{\omega}_k^m(H) = 1$. We use this average only in figure 5. The empirical surveillance metrics use the top-10 weights calibrated separately by side in table 2.

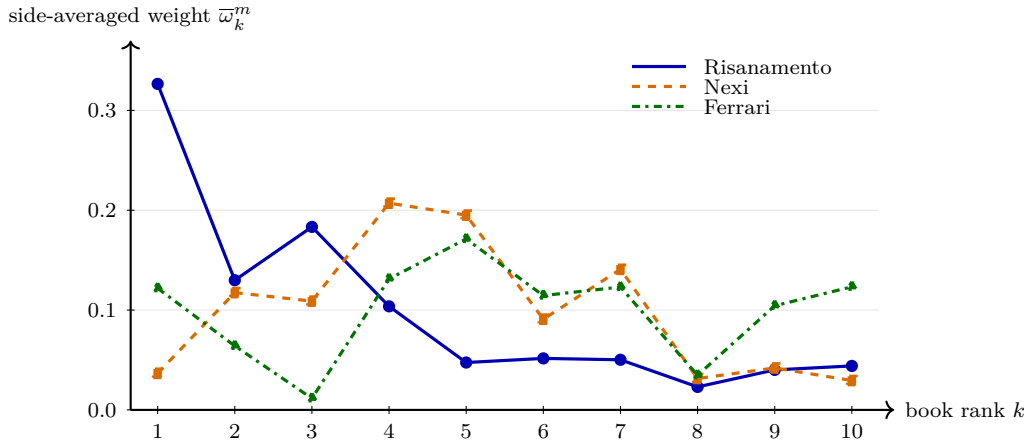


Figure 5: Empirical profiles of the top-10 depth kernel estimated on the full sample at $H = 10$ seconds. Each line is the arithmetic mean of the separately normalized bid and ask weights in equation (11). The mean permits one curve per instrument on a common rank axis and is used only for visualization. Surveillance metrics use the underlying side-specific kernels.

To expose the two empirical inputs rather than only their normalized product, Figures 6 and 7 plot the visibility profiles and the profiles of execution risk, each estimated separately by side used in the calibration. The second plot reports $\hat{p}_k^{m,s}(H)$ directly; the kernel uses its protection complement $\hat{\rho}_k^{m,s}(H) = 1 - \hat{p}_k^{m,s}(H)$.

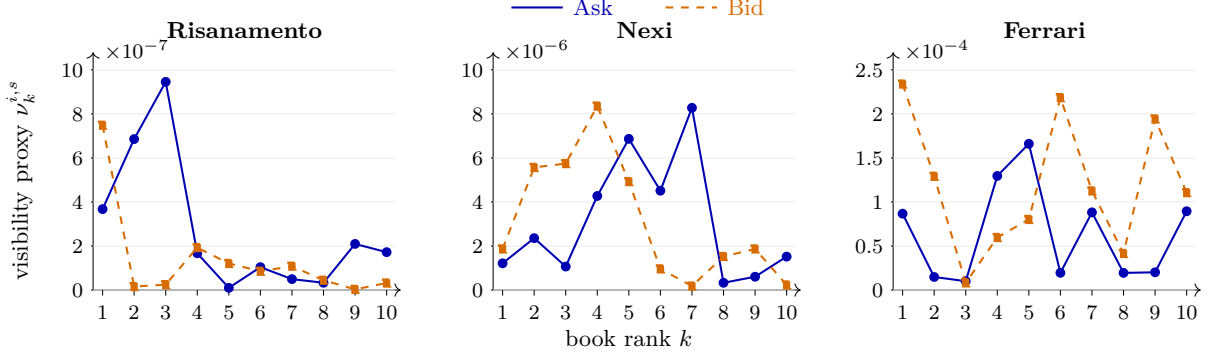


Figure 6: Side-specific empirical visibility proxy by book rank. Each panel shows the stored `visibility_component` values from the full-sample top-10, $h = 10$ -second calibration. Separate axis multipliers preserve the instrument-specific price units; vertical magnitudes should not be compared across instruments.

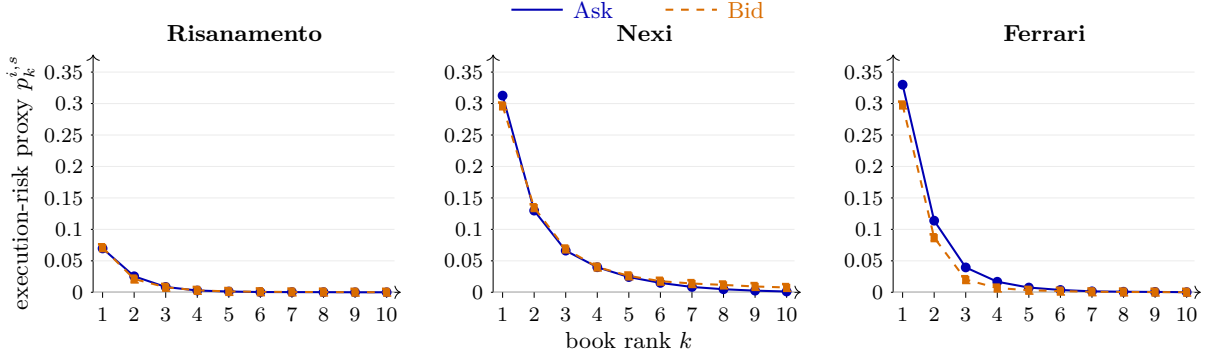


Figure 7: Side-specific empirical execution-risk proxy by book rank. The plotted quantity is the stored h -second reach probability $p_k^{i,s}$ (`hit_probability`) from the full-sample top-10, $h = 10$ -second calibration. The empirical kernel uses its protection complement $\rho_k^{i,s} = 1 - p_k^{i,s}$.

Using the calibrated kernel, we define participant i 's weighted top- n liquidity on side s as

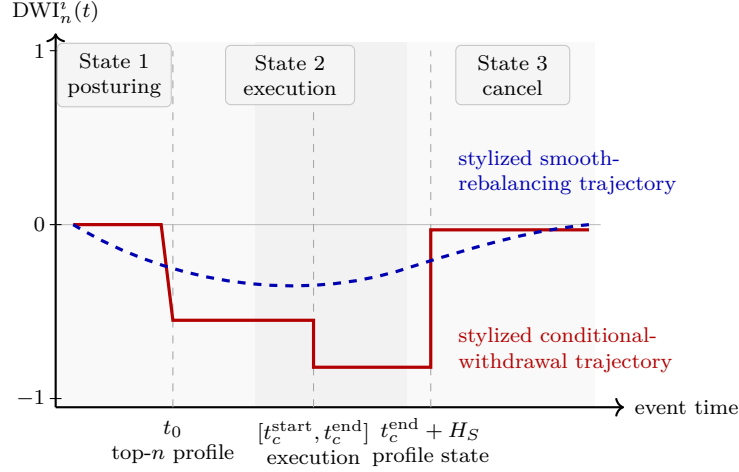
$$L_n^{i,s}(t) = \sum_{k=1}^n \omega_k^s(t) \tilde{V}_k^{i,s}(t). \quad (12)$$

We summarize the difference between the two sides with the *Multilevel Distance-Weighted Imbalance*:

$$\text{DWI}_n^i(t) = \begin{cases} \frac{L_n^{i,a}(t) - L_n^{i,b}(t)}{L_n^{i,a}(t) + L_n^{i,b}(t)}, & L_n^{i,a}(t) + L_n^{i,b}(t) > 0, \\ \text{undefined}, & L_n^{i,a}(t) + L_n^{i,b}(t) = 0. \end{cases} \quad (13)$$

When the denominator is positive, $\text{DWI}_n^i(t) > 0$ denotes an ask-heavy profile and $\text{DWI}_n^i(t) < 0$ denotes a bid-heavy profile. The denominator scales the difference by the participant's total weighted footprint, reducing sensitivity to the absolute scale of that footprint within the same instrument and kernel calibration. Values near $+1$ and -1 correspond to profiles concentrated almost entirely on the ask and bid sides, respectively. A value near zero denotes a balanced positive footprint, whereas the statistic remains undefined when both weighted sides are absent. The reconstructed state series additionally records [a single transition to a zero profile when the](#)

previously observed profile over the top n ranks disappears; DWI is set to zero only for that explicit operational state. Setting $n = 1$ gives the best-level measure. Larger values of n also capture candidate liquidity posted behind the touch. DWI is a descriptive diagnostic of profile shape. Neither its level nor its change defines candidate-set membership or the strict gate. Figure 8 illustrates two stylized paths; it is not an empirical classification rule.



For the opposite spoofing direction, the sign pattern is mirrored.

Figure 8: Stylized behavior of the multilevel distance-weighted imbalance around an execution. The trajectory with conditional withdrawal shows a large top- n imbalance returning rapidly toward zero after the execution on the opposite side. The smooth trajectory illustrates gradual rebalancing. Neither trajectory is an empirical classification rule.

5.1.3 Conditional Withdrawal Reconstruction

A static multilevel imbalance cannot establish manipulative intent. A market maker can also become temporarily imbalanced while controlling inventory or responding to adverse selection. The additional observable feature is whether the profile disappears conditionally after the participant's execution on the opposite side, as predicted by the stylized mechanism.

Let c denote an execution for canonical actor i . Its ordered constituent execution messages share the trading partition, actor key, order identifier, execution role, and side, and they satisfy the operational rule for the gap between execution messages. Passive executions also share the resting price. Aggressive executions use reported traded price and quantity and may span several price levels within the same order or sweep. The timestamps t_c^{start} and t_c^{end} denote the first and last constituent execution messages. The quantity Q_c is their total executed quantity, and $s_c \in \{b, a\}$ is the actor's execution side. We denote the side opposite the execution, on which the candidate posture is sought by $\bar{s}_c$. The observation times before and after the execution are defined from the reconstructed state series in source order. Each state is recorded after applying its message. Therefore, t_c^- denotes the last row in the actor's state series whose source index is strictly smaller than the first constituent execution message's index. Let $t_c^{\text{target}} = t_c^{\text{end}} + H_S$, where $H_S = 10$ seconds is the structural state horizon. The state after the execution t_c^+ is the last reconstructed state at or before t_c^{target} whose source index is no smaller than the final constituent execution message's index. SCI, the collapse measure for each side $C_{c,n}^{i,s}$, and MSCI

are undefined when either required reconstructed state is unavailable.

We also impose [an episode window before execution](#) $\Theta > 0$. [An order on the opposite side enters the candidate profile](#) for execution c only when its first observed posting time τ_o satisfies

$$0 \leq t_c^{\text{start}} - \tau_o \leq \Theta. \quad (14)$$

The empirical analysis uses $\Theta = 90$ seconds as an exploratory default. Let $\mathcal{P}_{c,n,\Theta}^{i,\bar{s}_c}$ contain the orders that are active immediately before the first constituent execution message, belong to actor i , lie on side $\bar{s}_c$ within the first n book ranks, and satisfy [equation \(14\)](#). The detector fixes this candidate set at the first constituent execution message. It records each order's visible quantity, rank, [contribution to relative depth after applying the kernel weights](#), and first-observed time. In particular, the raw candidate quantity is

$$Q_{c,n}^{i,\bar{s}_c} = \sum_{o \in \mathcal{P}_{c,n,\Theta}^{i,\bar{s}_c}} q_o(t_c^-). \quad (15)$$

The age restriction therefore governs [the attribution of candidate orders](#), matched withdrawal, WMSCI, and the strict event gate. It does not filter the state series used by SCI and MSCI.

The *Spoofing Conditionality Index* instead measures the absolute change in the normalized shape of actor i 's complete top- n resting profile:

$$\text{SCI}_{c,n}^i = \left| \text{DWI}_n^i(t_c^-) - \text{DWI}_n^i(t_c^+) \right|. \quad (16)$$

An absolute imbalance change does not identify which side lost liquidity. We therefore also measure the fraction of weighted liquidity that disappears from each side after the execution using an exact piecewise definition:

$$C_{c,n}^{i,s} = \begin{cases} \frac{[L_n^{i,s}(t_c^-) - L_n^{i,s}(t_c^+)]_+}{L_n^{i,s}(t_c^-)}, & L_n^{i,s}(t_c^-) > 0, \\ 0, & L_n^{i,s}(t_c^-) = 0. \end{cases} \quad s \in \{b, a\}. \quad (17)$$

where $[x]_+ = \max\{x, 0\}$. A large $C_{c,n}^{i,\bar{s}_c}$ indicates that the actor's top- n liquidity decreased on the side opposite to the execution. A two-sided quote refresh after a midprice change can instead generate material collapse on both sides. The boundary values have a direct interpretation: $C_{c,n}^{i,s} = 1$ means that the entire [weighted profile before execution](#) on side s disappeared, whereas $C_{c,n}^{i,s} = 0$ means that it did not decrease or was absent before execution.

To summarize these diagnostics while preserving evidence in both directions, define [the signed contrast at the event level](#)

$$\text{MSCI}_{c,n}^i = \frac{1}{2} \text{SCI}_{c,n}^i + C_{c,n}^{i,\bar{s}_c} - C_{c,n}^{i,s_c}. \quad (18)$$

Because DWI lies in $[-1, 1]$, $\text{SCI}/2$ and each collapse term lie in $[0, 1]$. Therefore, MSCI lies in $[-1, 2]$ without an additional clipping step. A positive MSCI means that normalized shape change and [collapse on the opposite side](#) outweigh [collapse on the same side](#). Zero is the neutral contrast, whereas a negative MSCI means that [collapse on the same side](#) dominates.

The positive part in the definition of C distinguishes liquidity disappearance from liquidity growth. It does not truncate [the subsequent contrast between sides](#). The three inputs remain separately inspectable, so a two-sided quote refresh is distinguishable from an asymmetric collapse. MSCI is [a secondary diagnostic at the event level that enters summaries of features for each actor](#). It is neither a probability nor matched-withdrawal evidence. It does not enter the four-condition event gate or the reported result tables. Operational outputs call it `MSCI_resting_profile` to emphasize that it uses [all active top- \$n\$ orders of the same actor, not only the candidate set restricted by age](#). Consequently, [balanced liquidity that has been resting for a long time](#) can dilute the measured shape change. MSCI should therefore be interpreted as context about the actor’s complete resting profile rather than as the collapse of the recent candidate layer.

5.1.4 Canonical Matched Withdrawal

A cancellation can be eligible for several nearby executions. To measure withdrawal without counting that cancellation more than once, let $\mathcal{E}(o)$ contain [the eligible executions of the same actor whose execution side is opposite to the cancelled order](#) that precede cancellation o by at most $\Delta\tau_W$, where $\Delta\tau_W = 2$ seconds in the reported specification. Eligibility is evaluated only when the cancelled order belongs to [the execution’s fixed candidate set defined before execution](#); the timing and [precedence in the source sequence](#) conditions stated below must also hold. We assign o deterministically to the canonical execution

$$a(o) = \arg \min_{c \in \mathcal{E}(o)} (t_o - t_c^{\text{end}}, t_c^{\text{start}}, \text{id}_c), \quad (19)$$

where the tuple is ordered lexicographically and the stable execution identifier breaks any final tie. Eligibility also requires [precedence in the source sequence](#): the cancellation follows the final constituent execution message in both event time and source sequence. For [eligible cancellations among orders in the top \$n\$ ranks](#), define $\mathcal{C}_{c,n}^i = \{o : a(o) = c\}$. A physical cancellation is identified by its trading partition, source index, and [identifier of the candidate order](#). We preserve every eligible link for audit, but each physical cancellation contributes to at most one execution’s metrics.

An order can grow after the execution but before its cancellation. Such growth was not part of [the candidate profile observed before execution](#) and must not increase the withdrawal attributed to that profile. For each assigned order $o \in \mathcal{C}_{c,n}^i$, let $q_{c,o}^{\text{pre}}$ denote its visible quantity immediately before the execution and let q_o^{cancel} denote its visible quantity immediately before cancellation. We define the attributed quantity as

$$\bar{q}_{c,o} = \min\{q_{c,o}^{\text{pre}}, q_o^{\text{cancel}}\}. \quad (20)$$

[The raw quantity observed at cancellation](#) remains available for audit. The cap ensures that only liquidity already present in the candidate layer contributes to matched withdrawal. The

total assigned withdrawal and its ratio to execution quantity are

$$U_{c,n}^i = \sum_{o \in \mathcal{C}_{c,n}^i} \bar{q}_{c,o}, \quad R_{c,n}^i = \frac{U_{c,n}^i}{Q_c}, \quad (21)$$

where $Q_c > 0$ is the quantity executed in the execution by construction. Using the candidate quantity after imposing the age restriction $Q_{c,n}^{i,\bar{s}_c}$ from equation (15), we report the fraction of that profile removed as

$$A_{c,n}^i = \begin{cases} \frac{U_{c,n}^i}{Q_{c,n}^{i,\bar{s}_c}}, & Q_{c,n}^{i,\bar{s}_c} > 0, \\ \text{undefined}, & Q_{c,n}^{i,\bar{s}_c} = 0. \end{cases} \quad (22)$$

The two ratios answer different questions. $R_{c,n}^i$ compares the attributed withdrawal with the executed quantity, whereas $A_{c,n}^i$ compares it with the candidate profile that was available to withdraw. Under the reconstructed lifecycle of each order and the unique assignment of each physical cancellation, the cap in equation (20) keeps $A_{c,n}^i$ in $[0, 1]$ whenever its denominator is positive.

For event prioritization, we also summarize the scale and speed of the attributed withdrawal. Define the withdrawal quantity weighted by delay

$$\tilde{U}_{c,n}^i = \sum_{o \in \mathcal{C}_{c,n}^i} \bar{q}_{c,o} \exp\left(-\frac{t_o - t_c^{\text{end}}}{\tau_w}\right), \quad \tau_w = 10 \text{ seconds}. \quad (23)$$

Within the 2-second assignment window, the exponential factor decreases with cancellation delay. The decay scale τ_w prioritizes temporal proximity; it is not an estimate of a causal response rate. The *Withdrawal Multilevel Spoofing Conditionality Index* for each event is

$$\text{WMSCI}_{c,n}^i = \begin{cases} \log\left(1 + \frac{Q_{c,n}^{i,\bar{s}_c}}{Q_c}\right) \log\left(1 + \frac{\tilde{U}_{c,n}^i}{Q_c}\right) A_{c,n}^i, & Q_{c,n}^{i,\bar{s}_c} > 0 \text{ and } \tilde{U}_{c,n}^i > 0, \\ 0, & Q_{c,n}^{i,\bar{s}_c} = 0 \text{ or } \tilde{U}_{c,n}^i = 0. \end{cases} \quad (24)$$

The first logarithmic term measures the candidate posture relative to the execution. The second measures the rapidly withdrawn quantity on the same scale, and $A_{c,n}^i$ measures the share of the candidate posture that is removed. Under the piecewise definition, WMSCI is zero when there is no positive contribution from the candidate layer or from attributed withdrawal after weighting by delay; this does not imply that the unrestricted resting profile was unchanged. WMSCI is non-negative and has no fixed upper bound. We use it as the primary metric for ordering candidate events for human review. It is neither a probability nor a calibrated measure of manipulative intent, and its magnitude should not be interpreted as a proportional increase in suspiciousness. The full descriptive gate instead requires $U_{c,n}^i > Q_c$ and the separate conditions on the price path; it does not apply a WMSCI threshold. For each actor-partition combination, we summarize repeated behavior through counts and withdrawal rates rather than aggregating WMSCI into an actor ranking.

5.1.5 Price-Response Diagnostics

The variables for matched withdrawal show whether an actor removes liquidity that it placed on the opposite side after its own execution. They do not show whether the price path is consistent with the hypothesized mechanism. Such a path first moves in the direction favorable to the execution and then partly reverses after the candidate layer is removed. We measure these two features with separate diagnostics of the price response. They are economic consistency checks, not causal estimates of manipulation.

Let $b(t)$ and $a(t)$ denote the best bid and ask, with corresponding displayed quantities $q_1^b(t)$ and $q_1^a(t)$. When both quotes and positive depth at the best quotes are available, the two proxies for the market price are the midprice $M_{\text{mid}}(t) = [a(t) + b(t)]/2$ and the queue microprice $M_{\text{micro}}(t) = [a(t)q_1^b(t) + b(t)q_1^a(t)]/[q_1^b(t) + q_1^a(t)]$ [12]; otherwise the corresponding diagnostic is missing. Let $M(t)$ denote either proxy, and let t_0 be the earliest first-observed posting time among the eligible candidate orders for execution c . We encode the execution direction as

$$h_c = \begin{cases} +1, & \text{if the execution is a sell execution,} \\ -1, & \text{if the execution is a buy execution.} \end{cases} \quad (25)$$

Under this convention, a positive signed change is favorable to the execution: a sell benefits from an upward move and a buy from a downward move. We measure the favorable movement between the posting of the candidate layer and execution as

$$\text{FPM}_c^i = h_c [M(t_c^-) - M(t_0)], \quad (26)$$

Reversion is anchored to each assigned cancellation rather than to the end of the execution. Let t_o denote the cancellation timestamp, t_o^- the last reconstructed state strictly before its source index, and $H_R = 2$ seconds the reversion horizon. Let $\mathcal{C}_{c,n}^{i,\text{obs}}$ contain the assigned cancellations for which the relevant pre-cancellation and horizon values of M are observed and $w_{c,o} > 0$ is finite. For each execution, reversion after cancellation is measured as a mean weighted by attributed quantity and cancellation delay

$$\text{REV}_c^i = \frac{\sum_{o \in \mathcal{C}_{c,n}^{i,\text{obs}}} w_{c,o} h_c [M(t_o^-) - M(t_o + H_R)]}{\sum_{o \in \mathcal{C}_{c,n}^{i,\text{obs}}} w_{c,o}}, \quad w_{c,o} = \bar{q}_{c,o} \exp\left(-\frac{t_o - t_c^{\text{end}}}{\tau_w}\right), \quad (27)$$

with $\tau_w = 10$ seconds in the reported run. Here $M(t_o + H_R)$ denotes the last reconstructed state at or before the target time, provided that the message sequence covers that target. The statistic is undefined when no assigned cancellation contributes a finite positive weight with both required price observations. A positive FPM_c^i means that the price moved in the direction favorable to the execution while the candidate layer was present. A positive REV_c^i means that the proxy for the market price moved against the favorable direction observed before execution after the assigned cancellation or cancellations. For execution price P_c , weighted by executed quantity, we also define the local execution advantage relative to the benchmark measured before the posture:

$$\text{ADV}_c^i = h_c [P_c - M(t_0)]. \quad (28)$$

The execution advantage is positive when a sell executes above the benchmark or a buy executes below it.

We compute each diagnostic with both the midprice and the queue microprice in [the records for each event](#). The reported summary tables and the event gate use the midprice versions. We report price responses separately from withdrawal counts, quantities, and rates instead of multiplying all components into one detector. Withdrawal variables describe the reconstructed message sequence. The variables FPM, REV, and ADV describe the associated price path. This separation distinguishes repeated conditional withdrawals without a favorable price response from episodes that also exhibit [favorable movement before execution and reversion after cancellation](#).

5.1.6 Event Reconstruction

We reconstruct events [from individual order messages rather than aggregate snapshots](#). A resting order can change book rank when the best quote moves even though its owner has neither modified nor cancelled it. Snapshot changes alone can therefore confound repricing of the surrounding book with withdrawal by the participant. The reconstruction follows the economic sequence from actor identification to execution, withdrawal, and price response:

1. Resolve the canonical actor and classify each execution message as exclusively passive or aggressive. Construct execution c from compatible constituent execution messages while preserving the ordered mapping to raw messages.
2. Reconstruct all active orders of actor i immediately before t_c^{start} , assign them to side and rank, and compute the unrestricted $L_n^{i,b}(t_c^-)$, $L_n^{i,a}(t_c^-)$, and $\text{DWI}_n^i(t_c^-)$.
3. [From the orders on the opposite side among the top \$n\$ ranks in the state before the execution](#), retain the fixed candidate set $\mathcal{P}_{c,n,\Theta}^{i,\bar{s}_c}$ whose first-observed times satisfy the age restriction.
4. Follow the complete actor state through the structural target $t_c^{\text{end}} + H_S$ and select t_c^+ according to [the rule based on source order and event time](#) above.
5. Retain every eligible execution–cancellation link. Assign each physical cancellation canonically using [equation \(19\)](#), exclude competing links from metric totals, and cap attributed quantity at [the visible quantity of the candidate order before execution](#) using [equation \(20\)](#).
6. Compute SCI, [collapse on each side](#), and MSCI from [the states of the unrestricted resting profile](#) at t_c^- and t_c^+ .
7. Report assigned withdrawal quantity, [ratio of withdrawal to execution](#), [fraction of the profile that was removed](#), and WMSCI. Use WMSCI to order [candidate events](#) for human review without applying it as [a threshold for the strict gate](#).
8. Compute signed price diagnostics from [the first posting in the candidate layer](#), through the execution, to [the window after cancellation](#).

9. Summarize repeated behavior with execution counts and [diagnostics of recurrence for each actor](#). Retain [raw constituent execution messages](#) as provenance.

This procedure is designed to distinguish ordinary book repricing from withdrawal of [candidate liquidity attributed to the participant](#). Canonical assignment ensures that one physical cancellation does not inflate the metrics of several neighboring executions under the stated eligibility rule.

At the event level, the gate for a *sequence compatible with spoofing* passes only when four conditions hold simultaneously: (i) [cancellation of at least one eligible order on the opposite side](#) is canonically assigned within 2 seconds of the execution; (ii) the finite attributed withdrawal exceeds [the positive quantity executed in the execution](#), $U_{c,n}^i > Q_c$; (iii) the observed midprice FPM_c^i is strictly positive; and (iv) [the observed midprice \$REV_c^i\$ measured from cancellation](#) is strictly positive. [A price diagnostic that is missing or right censored](#) fails the conjunction. This transparent rule supports analyst triage; it is descriptive and is not a statistical test, manipulation label, or proof of intent.

The reported specification uses a 10-second window for structural reconstruction, a 2-second window for both rapid attribution and reversion measured from cancellation, and a maximum age of 90 seconds for candidate orders. We estimate the empirical top-10 kernel separately for each instrument. [The components describing withdrawal and the price response](#) remain separate rather than being reduced to [a single composite score](#).

6 Results

6.1 Cross-Instrument Detection Summary

[Rapid withdrawals by the same actor](#) occur in all three samples and after both passive and aggressive executions. The complete strict sequence is necessarily a smaller subset because, in addition to rapid canonical matching, it requires $U_{c,n}^i > Q_c$, [positive favorable movement before execution](#), and [positive reversion measured from cancellation](#). Table 3 reports this narrowing while using the same top-10 state for each actor and the same reconstruction of candidate orders with the age restriction. Each instrument uses its empirical kernel estimated on the full sample from [table 2](#) with kernel horizon $H = 10$ seconds.

A matched execution is a passive or aggressive execution followed within 2 seconds by a canonically assigned cancellation of [a candidate order placed by the same actor on the opposite side](#). The separate 10-second interval is [the horizon used for structural reconstruction](#). [The quantity attributed to the cancellation](#) cannot exceed the candidate order’s visible quantity before execution, while [the raw quantity observed at cancellation](#) remains available for audit. The table distinguishes executions from [their raw constituent execution messages](#) and counts each assigned physical cancellation once. [The column for firm fallback](#) records the coarser cases in which the original client code is missing. The strict column then retains matched executions that also satisfy [the condition on withdrawal size and the two conditions on the price response](#). These quantities are descriptive screening diagnostics, not manipulation labels or evidence of intent.

Table 3: Actor-attributed execution populations and reconstructed withdrawals by execution anchor. Individual execution messages provide provenance; executions are the analytical unit. Firm fallback counts matched executions for which the client code is missing and the firm identifier defines the actor. Assigned cancellations are counted once. Strict is the four-condition spoofing-compatible sequence gate.

Instrument	Anchor	Executions	Execution messages	Matched	Firm fallback	Assigned cancels	Strict
FERRARI	Aggressive	4,333	9,335	1,153	1,131	1,760	165
FERRARI	Passive	8,664	9,329	1,322	1,147	2,009	205
NEXI	Aggressive	3,849	7,425	620	576	944	100
NEXI	Passive	6,663	7,424	1,064	791	1,867	131
RISANAMENTO	Aggressive	9,956	16,029	71	44	84	10
RISANAMENTO	Passive	15,016	15,929	134	114	136	2

The main empirical pattern is a gap between mechanical matching and the complete price-consistent sequence. The reconstruction identifies 2,475 matched executions for Ferrari, 1,684 for Nexi, and 205 for Risanamento. The strict gate retains 370, 231, and 12, respectively. Favorable pre-execution movement is positive in 54.9%, 52.8%, and 56.6% of observed matched executions, whereas positive reversion occurs in 35.9%, 36.5%, and 21.7%. Thus, the withdrawal condition is not being used as a substitute for the full hypothesized price path. These are marginal shares among [observed diagnostics for matched executions](#) and do not decompose [attrition from the strict set](#) across the jointly required conditions.

These empirical quantities are direct reconstruction counts and conditional descriptive shares. No counterfactual comparison or inferential test is applied. The table therefore does not provide p -values, confidence intervals, or evidence that withdrawal is more likely after the focal actor’s own execution than under an otherwise comparable no-execution condition.

Event-accounting denominators. Table 3 reports [the provenance of raw executions, the composition of execution anchors, and counts that use firm fallback](#), and uniquely assigned physical cancellations for each instrument. Raw execution messages provide provenance, whereas executions are the analytical unit for all conditional shares.

6.2 Risanamento

Risanamento has the smallest matched set, and only 12 executions satisfy the strict four-condition gate. Favorable pre-execution movement is nevertheless positive in 56.6% of matched executions with an observed diagnostic. Among matched executions with an observed reversion diagnostic, 21.7% have [positive reversion measured from cancellation](#). The strict gate additionally requires [positive favorable movement before execution](#) and attributed withdrawal exceeding execution quantity, so these marginal shares do not identify which requirement excludes any particular matched execution.

For each of the 205 matched executions, the review record preserves [a dossier organized around the execution and a reconstructed event window covering the top 10 ranks](#). These records organize observed facts, counter-evidence, and legitimate alternatives. They are triage aids, not empirical labels or proof of manipulation.

6.3 Nexi

Nexi exhibits matched withdrawals after both execution routes, but [the full path consistent with the price conditions](#) remains a subset. Reconstruction identifies 1,684 matched executions, of which 231 pass the strict gate. Positive [favorable movement before execution](#) occurs in 52.8% of matched executions with an observed diagnostic, while [positive reversion measured from cancellation](#) occurs in 36.5%.

Differences from the other instruments are not causal contrasts because the samples cover different periods and actor compositions.

6.4 Ferrari

Ferrari has the largest matched set, with contributions from both passive and aggressive execution anchors. Reconstruction identifies 2,475 matched executions, and 370 pass the strict gate. Among [observed diagnostics for matched executions](#), the shares with [positive favorable movement before execution](#) and positive reversion are 54.9% and 35.9%.

These are descriptive screening outcomes, not manipulation labels or prevalence estimates.

6.5 Actor-Level Attribution and Interpretation

Recurrence is informative only when actor identity and execution role remain explicit. [The largest groups defined by actor and execution anchor differ in identity resolution, number of matched executions, withdrawal scale, and composition of the price path.](#) Table 4 therefore reports the three largest groups in each instrument [by number of matched executions](#) rather than collapsing them into one actor score. An actor is either an original client code or, only when that code is missing, a firm identifier. W/E maximum is the largest matched [ratio of withdrawal to execution](#) within the group; FPM+ and REV+ are the shares of matched executions with positive midprice favorable movement and reversion.

Table 4: Largest actor–anchor rapid-withdrawal groups by matched-execution count. Client denotes the original client code. Firm fallback denotes aggregation by firm only when that client code is missing. W/E max is the largest assigned withdrawal-to-execution ratio in the group. FPM+ and REV+ are descriptive shares among matched executions with finite midprice diagnostics.

Instrument	Actor	Identity	Anchor	Matched	W/E max	FPM+	REV+
FERRARI	157922_3	Firm fallback	Aggressive	367	60.0	43.6%	32.7%
FERRARI	157922_3	Firm fallback	Passive	257	77.5	47.1%	30.4%
FERRARI	619575_1	Firm fallback	Passive	242	34.5	52.1%	41.7%
NEXI	23523	Client	Passive	248	1200.0	64.5%	25.4%
NEXI	354310	Firm fallback	Passive	233	565.2	59.7%	28.9%
NEXI	628912	Firm fallback	Aggressive	191	1000.0	33.0%	57.1%
RISANAMENTO	628912	Firm fallback	Passive	83	80.0	79.5%	1.2%
RISANAMENTO	625922	Firm fallback	Aggressive	30	26.8	36.7%	56.7%
RISANAMENTO	17295	Client	Aggressive	22	44.4	36.4%	61.9%

The dispersion of FPM+ and REV+ across these groups shows why recurrence and price response must remain separate. A group can contain many matched withdrawals without exhibiting the same reversion pattern as another group. Because firm fallback can combine several

clients, a count at firm level has coarser attribution than a count at client level and should not be interpreted as the history of a single account.

The primary empirical object is the association, attributed to each actor, between execution and rapid withdrawal on the opposite side. Attribution is at client level when an original client code is observed and at the coarser firm level otherwise. We report its magnitude and repetition descriptively. WMSCI orders candidate events for review; MSCI remains a diagnostic that provides context from the complete resting profile; neither metric is a threshold for the strict gate or a score that ranks actors. An analyst must also consider the withdrawn order’s execution risk and distance from the touch, bilateral activity, inventory context, price response, and execution advantage. Together, these diagnostics support assessment of consistency with the stylized mechanism while retaining legitimate liquidity provision as an alternative explanation. Neither WMSCI nor a dashboard review category is treated as ground truth.

6.6 Coverage of Externally Supplied Surveillance Alerts

At the scope of the subject reported externally, the comparison recovers an execution in every distinct timed interval, but the complete behavioral sequence appears in only four intervals. The 36 timed source records become 34 distinct intervals after overlapping or endpoint-sharing records for the same instrument and external subject are merged. All 34 intervals contain raw subject activity and at least one recovered execution, while 30 contain an attributed withdrawal. Table 5 reports the corresponding hierarchy by instrument.

Table 5: Coverage of externally supplied timed alert records. Overlapping or endpoint-sharing records are merged only within the same instrument and external subject. Execution counts intervals containing at least one execution with a constituent execution message inside the closed union interval. Withdrawal and strict-scope columns count intervals containing at least one execution with the corresponding diagnostic at the event level. Strict subject scope requires at least one execution in the interval to satisfy all four gate conditions. The final column reports strict overlaps among intervals whose external identity maps to one canonical actor at the same granularity; entries report strict intervals divided by intervals with aligned identity granularity. The date-only Risanamento record is excluded from all timed denominators.

Instrument	Timed source records	Union intervals	Execution	Withdrawal	Strict scope	Strict / aligned intervals
Ferrari	35	33	33	30	4	0/3
Nexi	1	1	1	0	0	0/1
Total	36	34	34	30	4	0/4

All four strict overlaps at subject scope occur within the 30 Ferrari intervals for which identity granularity is not aligned and the main externally reported firm maps to two canonical actor keys. Correspondence with an exact actor is therefore not identifiable in those intervals. By contrast, the comparison contains four intervals with aligned identity granularity—three for Ferrari and one for Nexi—and none contains a strict sequence. The Nexi interval nevertheless contains one aggressive execution formed from four constituent execution messages, but it has no attributed withdrawal under the stated timing rule.

The Risanamento source identifies only one trading date. On that date, the reported subject has 59 reconstructed executions, including one execution with an attributed withdrawal and no

strict sequence. These counts over the full day establish that the subject is observable, but they cannot identify correspondence with the specific external event.

These results measure recovery relative to externally reported activity. Strict recall for an exact actor is identifiable only for the four unioned timed intervals with aligned identity granularity, none of which contains a strict sequence. The four strict overlaps among the remaining intervals are overlaps at subject scope, not recall for an exact actor. Because the source contains only positive cases and was not independently adjudicated again for manipulative intent, these quantities do not estimate recall or precision against true manipulation. No negative windows are available to estimate false positives. Moreover, the external alert rule and the strict four-condition gate need not define the same event population. The sparse strict overlap records limited agreement under two operational definitions. It cannot, by itself, identify false negatives, false positives, or the source of disagreement.

7 Discussion and Conclusion

The main empirical finding is that rapid withdrawal by the same actor is broader than the complete sequence compatible with spoofing. Matched episodes occur after both passive and aggressive executions in all three samples. Only a subset also exhibits the required withdrawal size, favorable movement before execution, and reversion after cancellation. This distinction matters because it prevents a high cancellation rate or a large withdrawal from standing in for the full mechanism.

The coverage audit based on external alerts reinforces this separation. The reconstruction covers execution in all 34 distinct timed intervals and attributed withdrawal in 30, whereas only four intervals contain a strict sequence at subject scope. None of the four intervals with aligned identity granularity contains a strict sequence, and the four strict overlaps at subject scope cannot be resolved to one canonical actor. Thus, the strict gate is not a surrogate for the external surveillance rule.

The mathematical construction makes the separation explicit. The quantities $U_{c,n}^i$, $R_{c,n}^i$, and $A_{c,n}^i$ describe how much of the candidate layer observed before execution can be attributed to the withdrawal and how its scale compares with the execution. The quantities FPM_c^i and REV_c^i describe the associated price path. Requiring both types of evidence produces the strict gate, but the conjunction remains a descriptive compatibility rule. It neither estimates spoofing prevalence nor establishes that any retained sequence was manipulative. WMSCI separately prioritizes review at the event level by combining the scale of the candidate layer, rapid attributed withdrawal, and the fraction of the profile that was removed. Unlike MSCI, it is not a measure of collapse in the complete resting profile; unlike the strict gate, it is not a pass-fail condition. MSCI is a separate diagnostic of the resting profile: it uses all active top- n orders of the same actor, so balanced liquidity that has been resting for a long time can attenuate the measured shape change. It therefore does not establish that the recent candidate layer itself collapsed.

The strongest limitation is the absence of a counterfactual comparison and an independently adjudicated label set. The analysis does not test whether withdrawal is more likely after the focal actor’s execution than after an otherwise comparable non-event, and repeated observations from the same actors or trading periods need not be independent. The external alerts add positive

reference periods but no negative controls. They support [recovery relative to the source alerts and accounting of strict recall for an exact actor under the audit’s rule for identity alignment](#), but they cannot identify recall or precision against independently adjudicated manipulation labels or calibrate detector thresholds. The depth kernels are also calibrated on the same full samples to which they are applied, so their [associations with the displayed price](#) are descriptive rather than [evidence from an out-of-sample evaluation](#). Reversion outcomes affected by right censoring fail the event gate, and the reported results do not test sensitivity to [the 100-millisecond threshold used to group constituent execution messages](#), the 90-second candidate age, or the reported 2- and 10-second windows.

Identity and sampling impose further limits. Firm fallback loses [resolution among clients within a firm](#) and can pool several accounts, so [recurrence measured at firm level](#) is not directly comparable with recurrence under a client code. Aggressive execution is [an observation of execution role](#), not evidence about the intent of the surrounding resting orders; ordinary liquidity taking remains an alternative explanation. [External alerts reported at firm level](#) can also span several internal actor keys, which prevents [evaluation at the level of an exact actor](#) in those intervals. The three instrument samples cover different periods and actor compositions, which rules out [causal comparisons across instruments](#).

The contribution is therefore an auditable surveillance workflow rather than an enforcement classifier. It preserves raw message provenance, separates withdrawal from [evidence about the price response](#), and exposes the gate conditions used for triage. Prospective use would require [a counterfactual design and thresholds specified in advance](#), kernel calibration on training periods only, [out-of-sample validation performed chronologically](#), and evaluation against independently reviewed cases with explicit [costs of false positives and false negatives](#).

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